

Limits of Black-Box Constructions of Pseudorandom Codes

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About Me

I'm from **Kerala**, India.

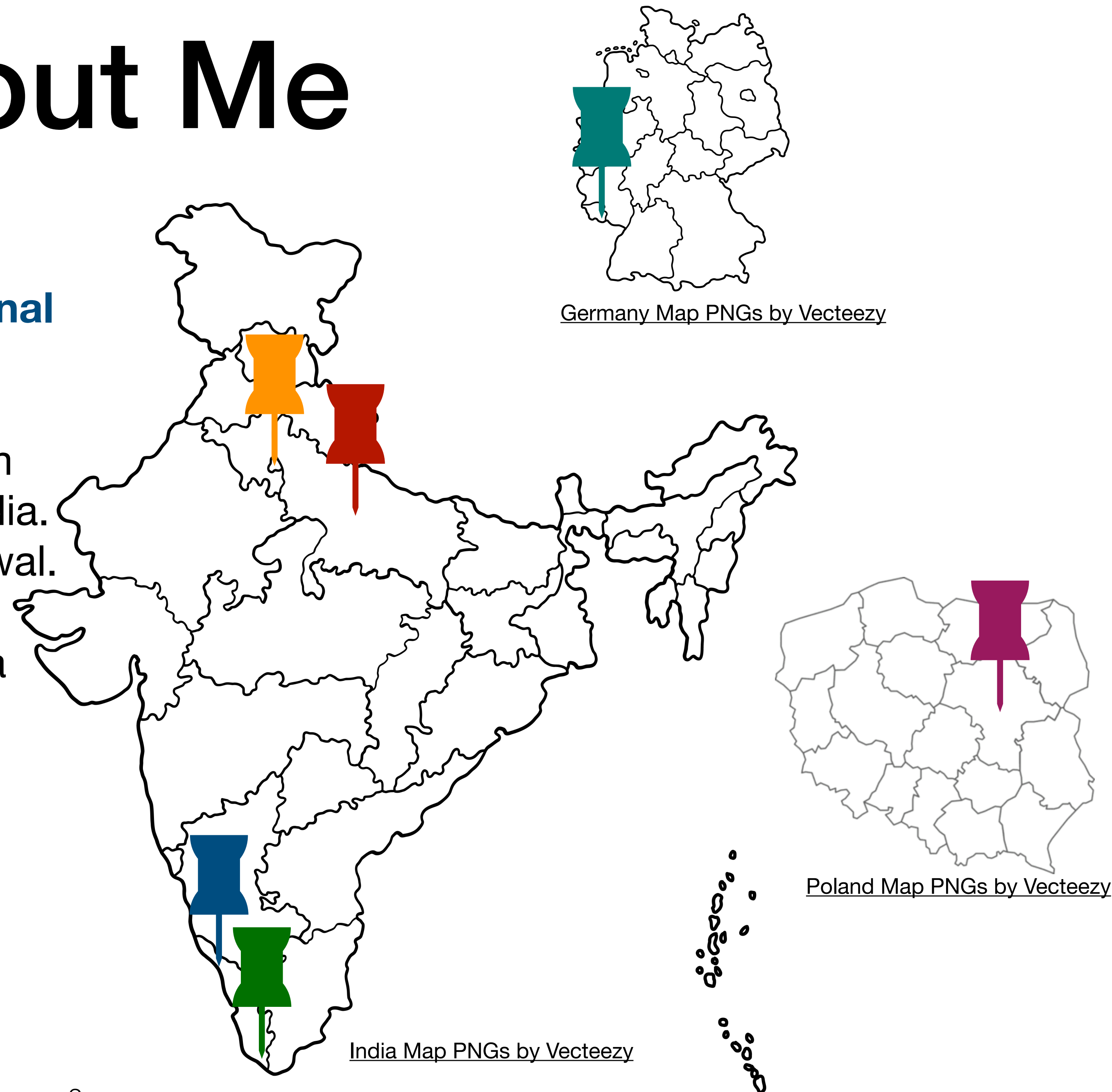
Bachelor's in Computer Science from **National Institute of Technology Calicut**, India.

Master's and PhD in Computer Science from **Indian Institute of Technology Kanpur**, India.
Under the guidance of Prof. Manindra Agrawal.

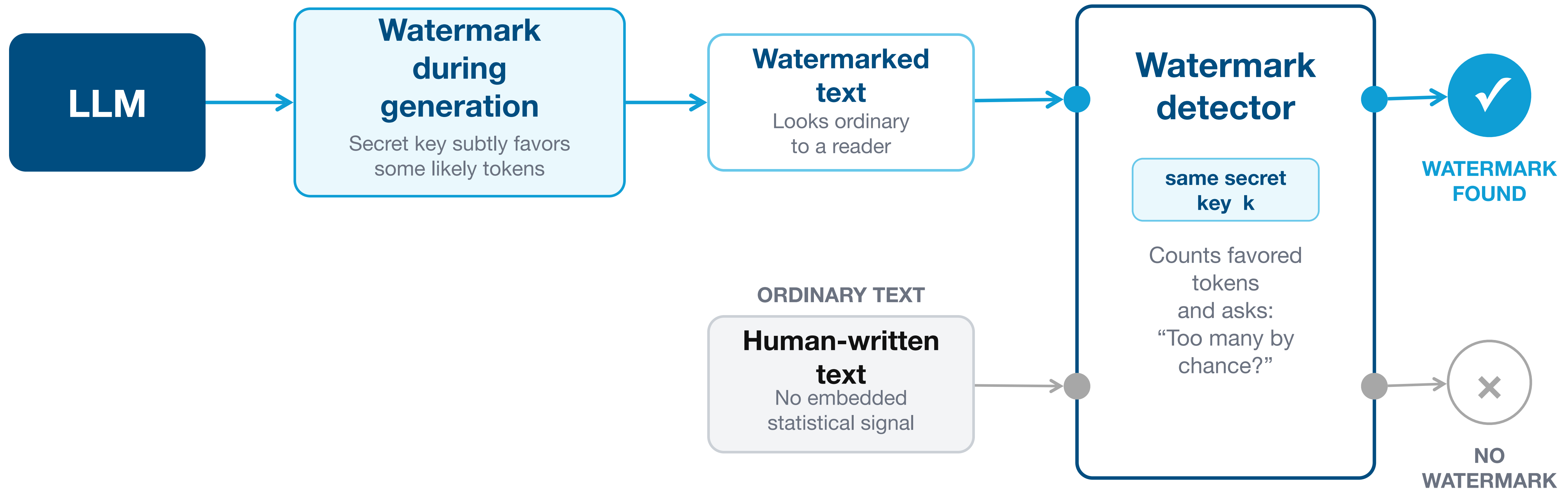
1st Postdoc at **IIT Delhi** under Prof. Venkata Koppula.

Currently a Postdoc at **CIPSA**, Germany.
Hosted by Prof. Nico Döttling.

Joining the **University of Warsaw** as an Assistant Professor.



Watermarking LLMs

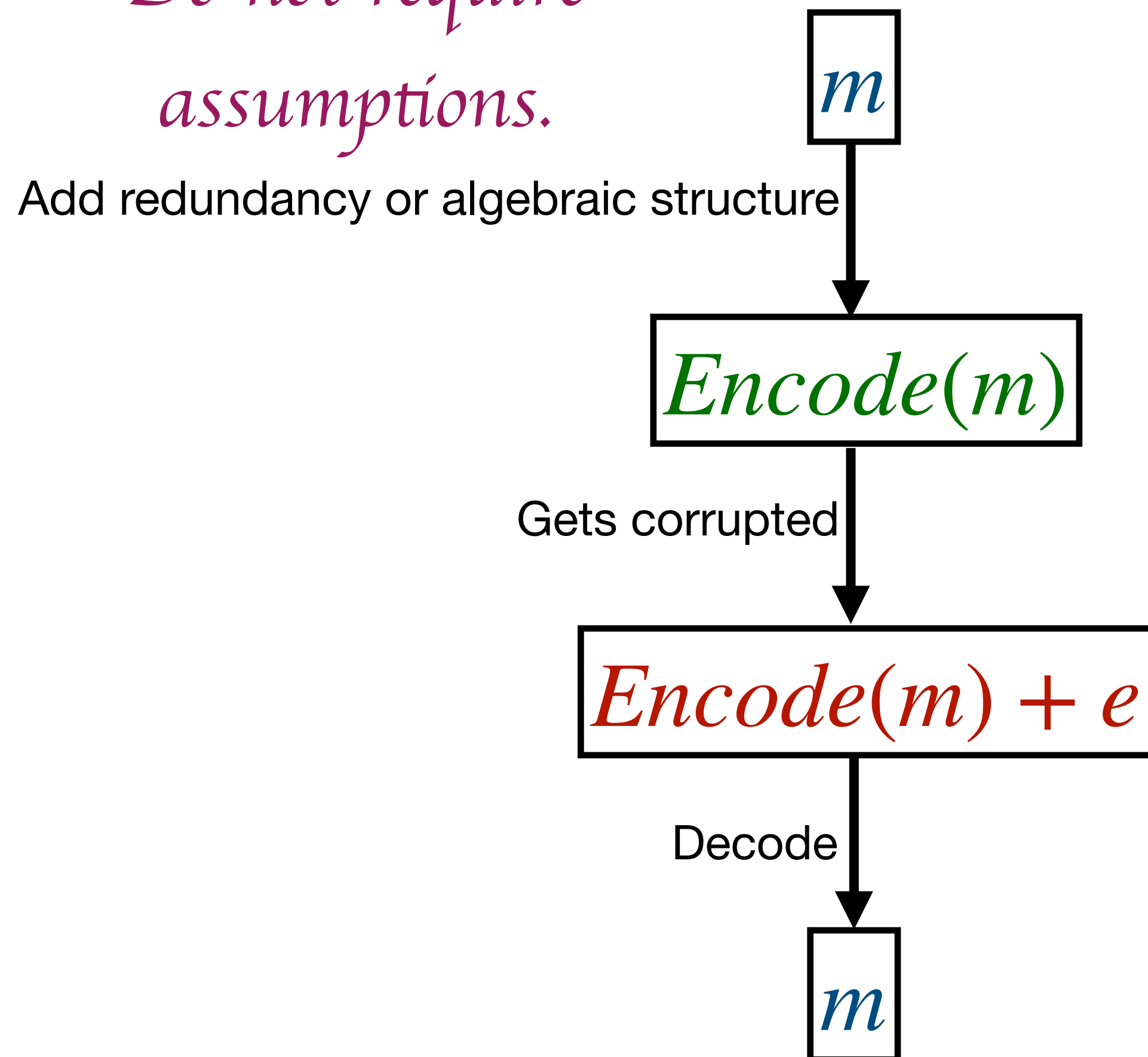


Pseudorandom Codes

Pseudorandomness Meets Error-Correction

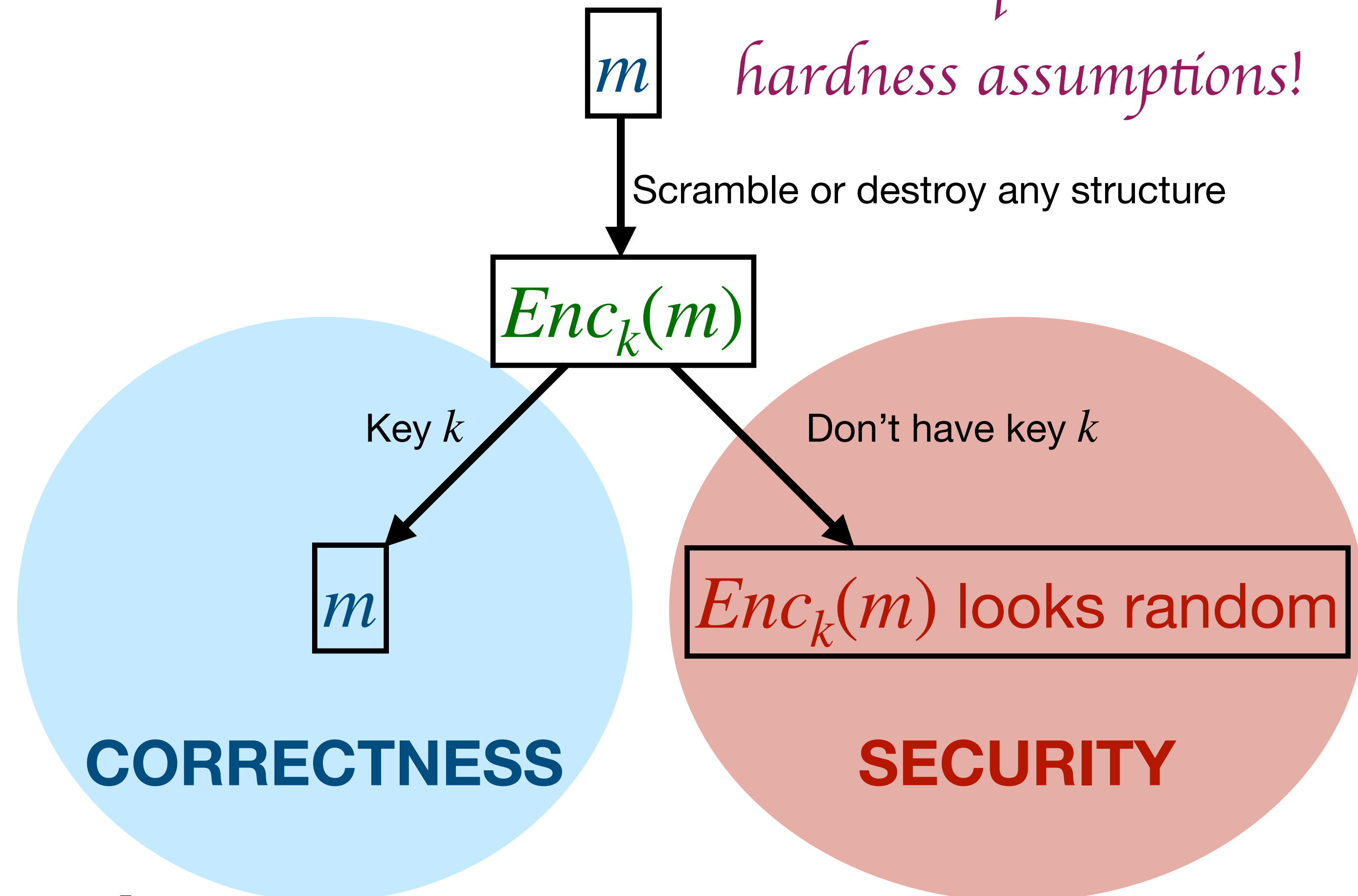
Coding Theory

Do not require assumptions.



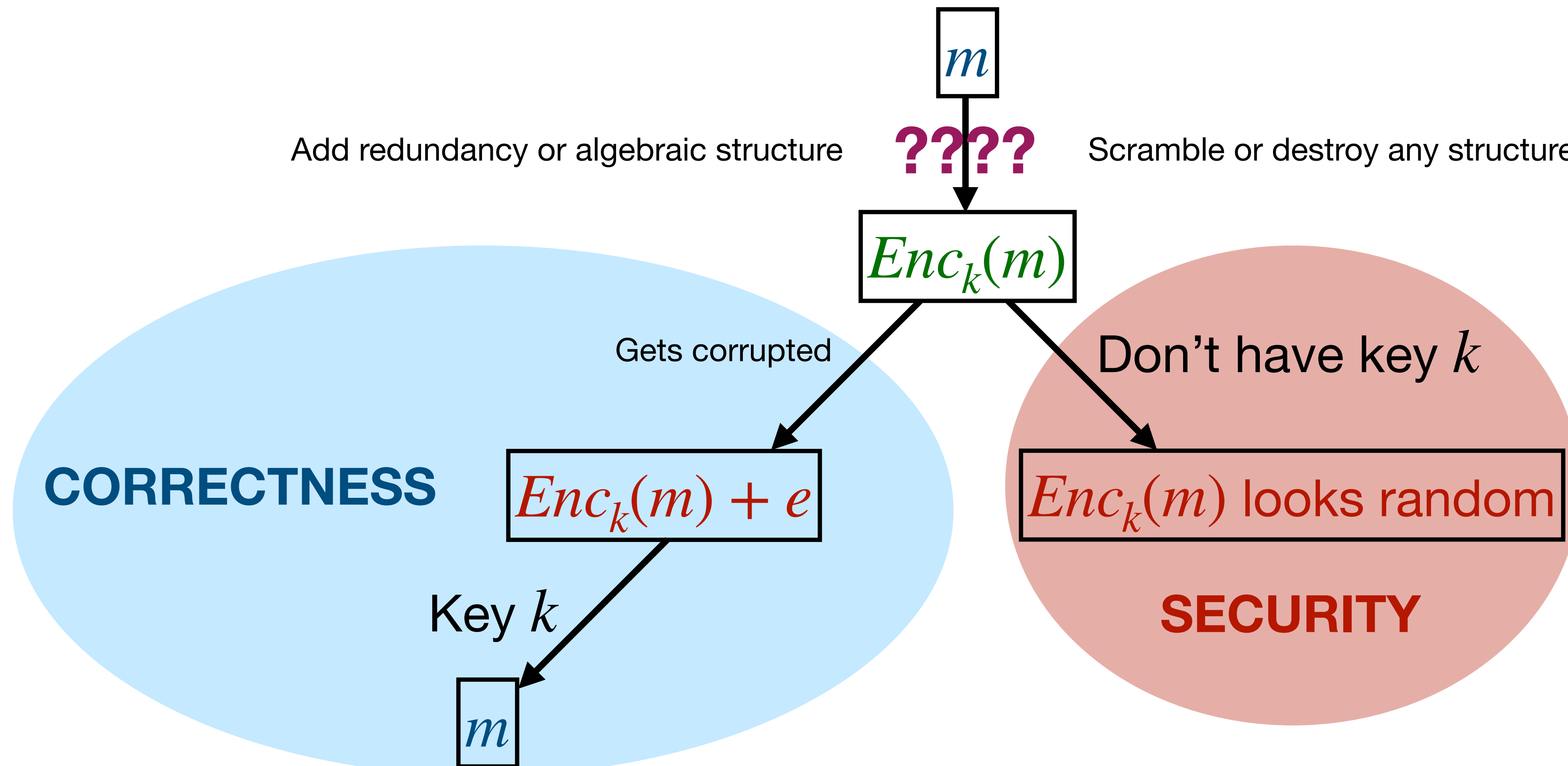
Cryptography

Requires hardness assumptions!



Pseudorandomness Meets Error-Correction

Pseudorandom Code



Initial Works on PRCs

Subexponential LPN

Planted Hyperloop and
Planted XOR

- Introduced by Christ and Gunn at **CRYPTO'24** for watermarking LLM/AI models.
- More works on constructions appeared in **NeurIPS'25, RANDOM'25**.
- In **FOCS'25** and **STOC'26**, extended to stronger settings.
- In **TCC'25**, two independent works showed strong barriers in building these !!!

Permuted Codes

Döttling, Müller, Rajasree
Garg, Gunn, Wang

Constructions

Syntax of PK-PRC

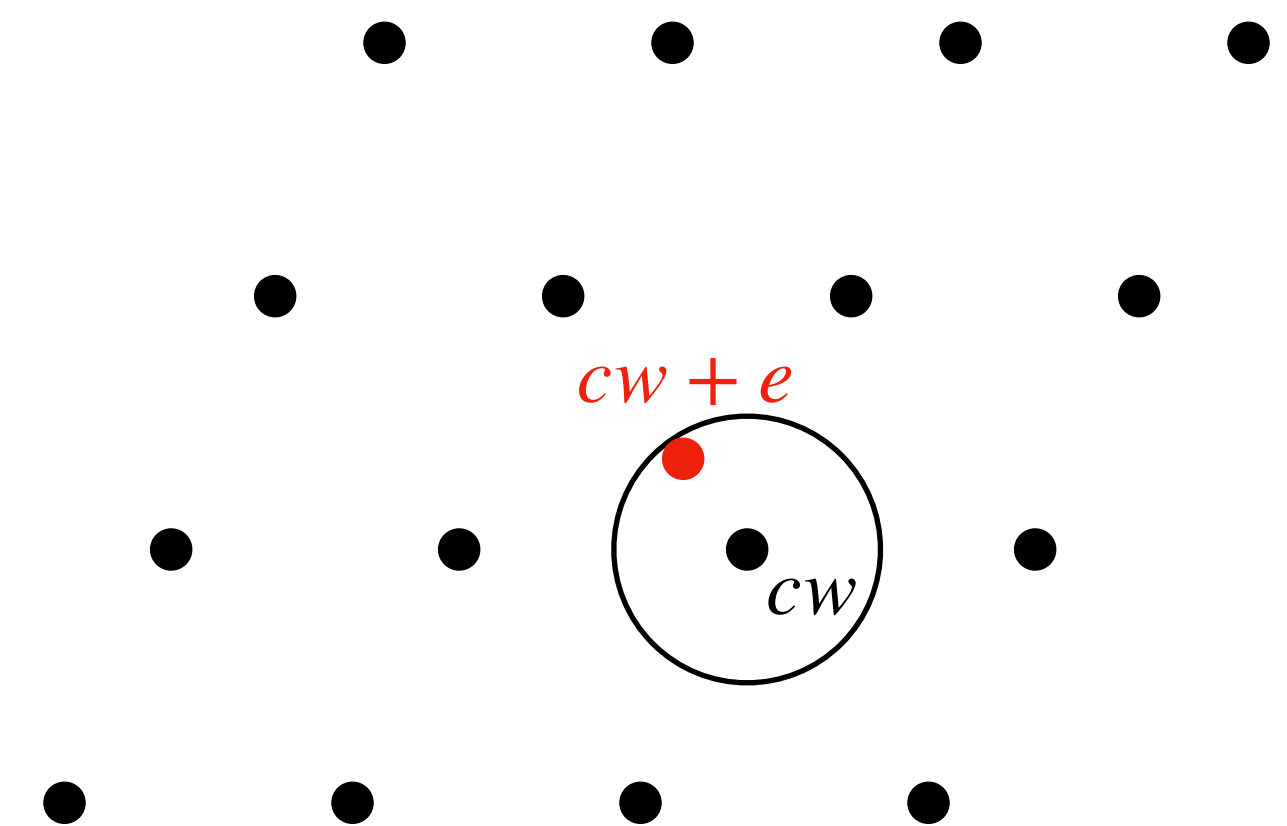
$$(\textcolor{brown}{pk}, \textcolor{blue}{sk}) \leftarrow \textit{Setup}(1^\lambda)$$

$$\textcolor{violet}{cw} \leftarrow \textit{Enc}(\textcolor{brown}{pk}, m)$$

$$\textcolor{violet}{cw} \approx \textcolor{blue}{uniform}$$

Pseudorandomness

$$m \leftarrow \textit{Dec}(\textcolor{blue}{sk}, \textcolor{violet}{cw} + \textcolor{red}{e})$$



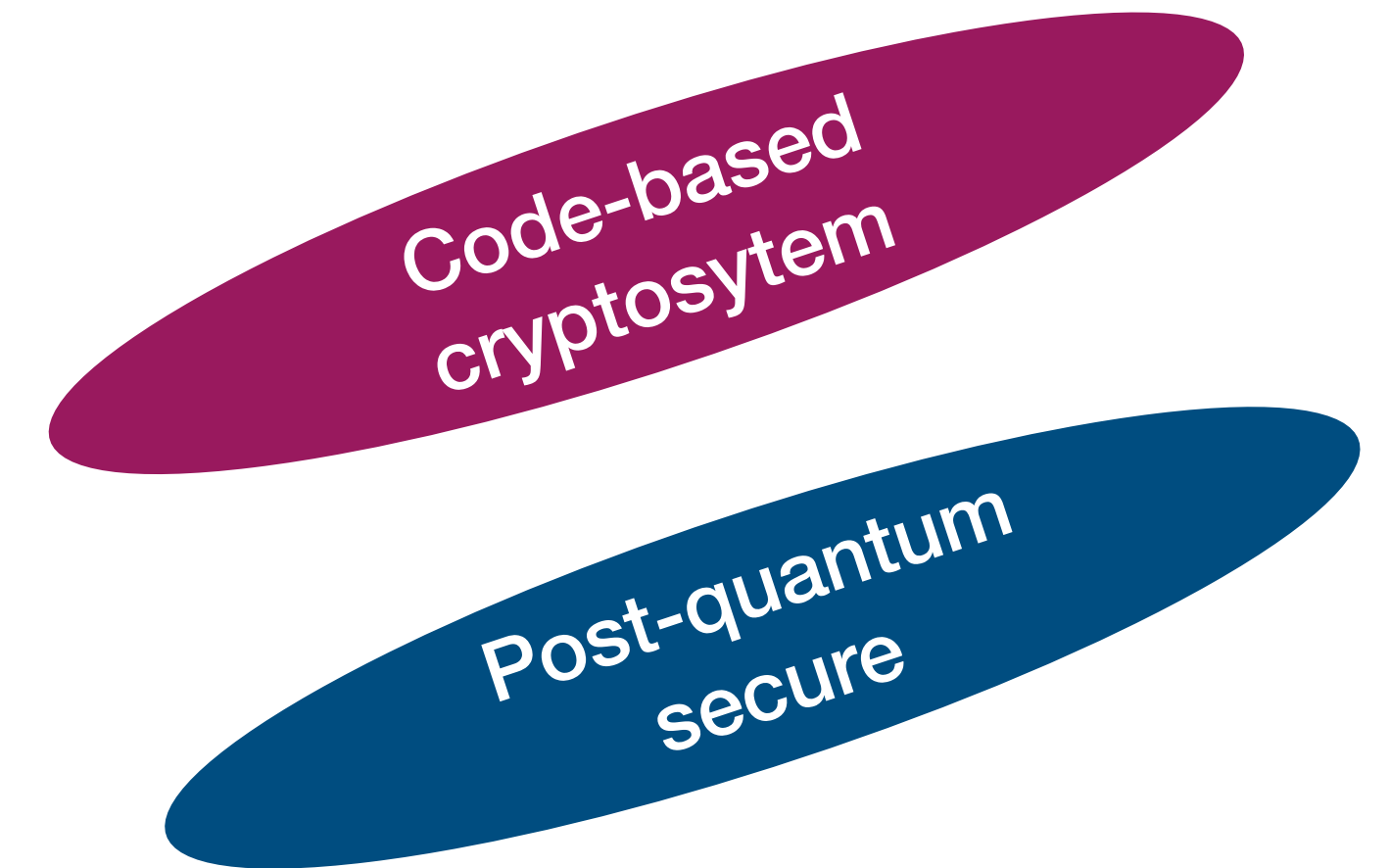
Robustness

McEliece Assumption

McE'78,DDMN'12

- $(G, td) \leftarrow$ generator-matrix of binary linear code C (Binary Goppa codes)
- $P \leftarrow$ permutation matrix
- $T \leftarrow$ invertible matrix

$$PGT \approx \textit{uniform}$$



- $pk = PGT, sk = (P, T, td)$
- Encryption is $pk \cdot m + e$ where $\textit{Ham}(e) = O(n/\log n)$.
- Decryption is $m' = \textit{Decode}(td, P^{-1} \cdot ct)$ and output $T^{-1}m'$

McEliece Assumption

- Ciphertexts are not random.
- Can handle only $O(n/\log n)$ errors rather than constant fraction.
- Many natural replacements for Goppa codes are **insecure** due to algebraic distinguishers.
- Known attacks fail if PGT is masked with **suitable amount of errors**.

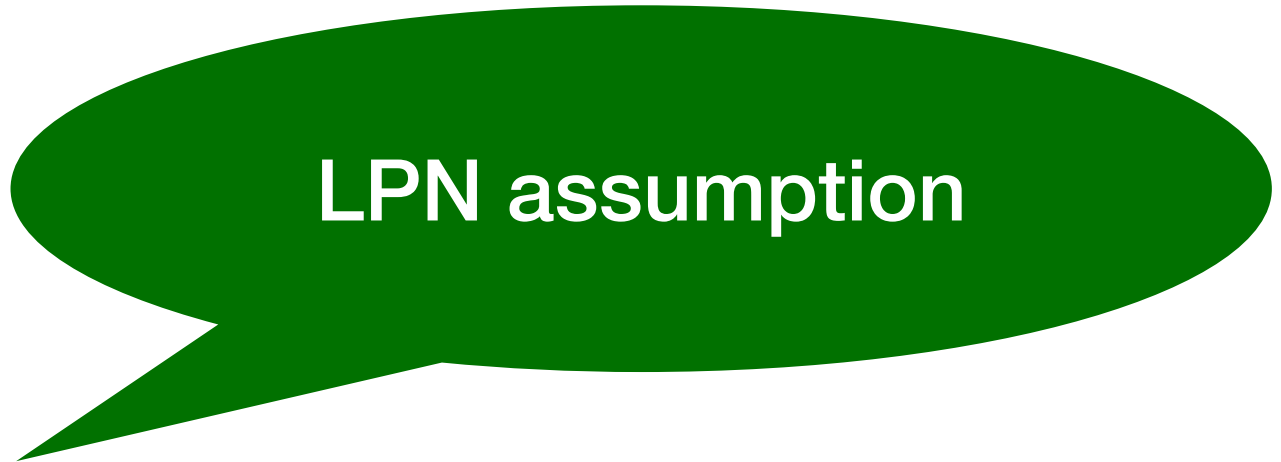
Noisy McEliece Assumption

DJKRW'26

- $G \leftarrow$ generator-matrix of binary linear code C (Algebraic-Geometric codes)
- $P \leftarrow$ permutation matrix
- $T \leftarrow$ invertible matrix
- $E \leftarrow \text{Ber}_\rho$

$$PGT + E \approx \textit{uniform}$$

- $pk = PGT + E$, $sk = (G, P, T)$
- Encryption is $pk \cdot m + e$ where $\textit{Ham}(e) = O(n)$.



LPN assumption

- Decryption is $m' = \textit{Decode}(P^{-1} \cdot ct)$ and output $T^{-1}m'$

Syntax of SK-PRC

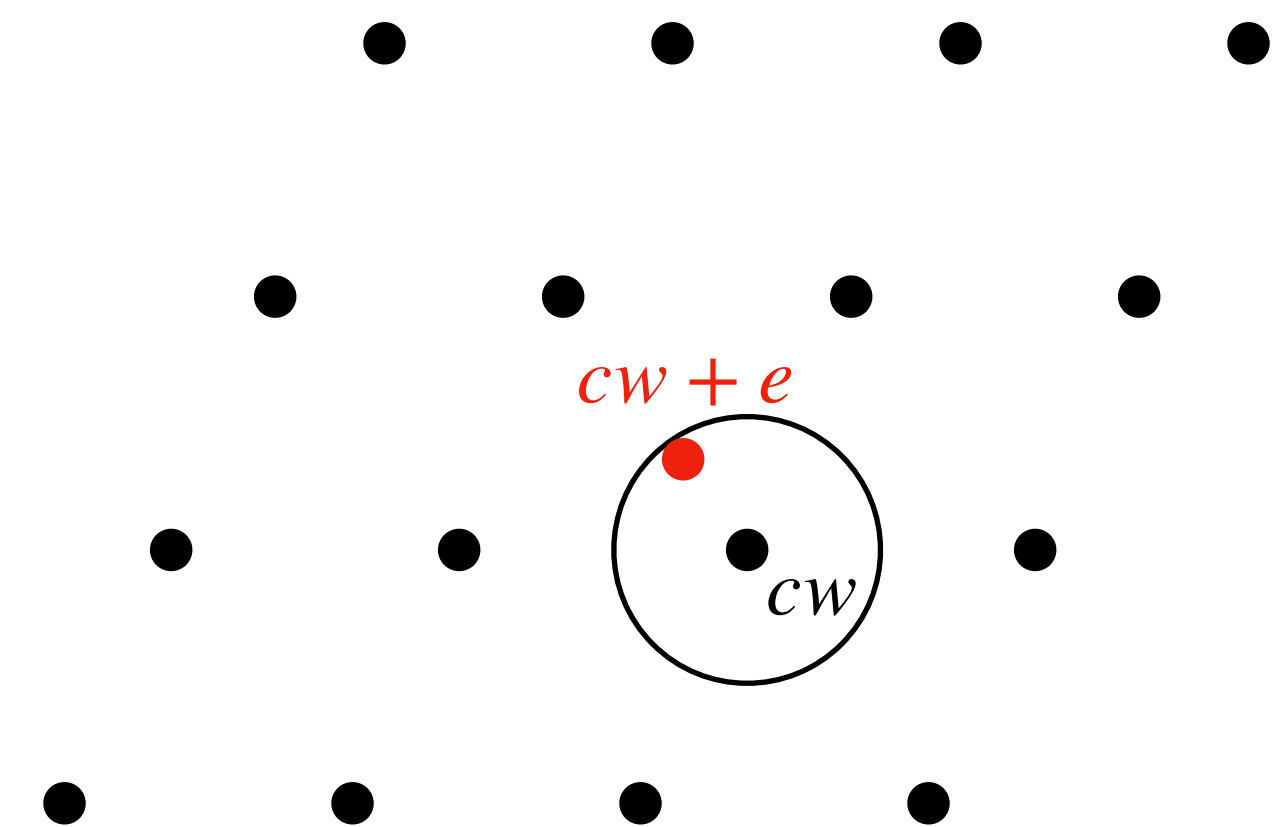
$$sk \leftarrow Setup(1^\lambda)$$

$$cw \leftarrow Enc(sk, m)$$

$$cw \approx uniform$$

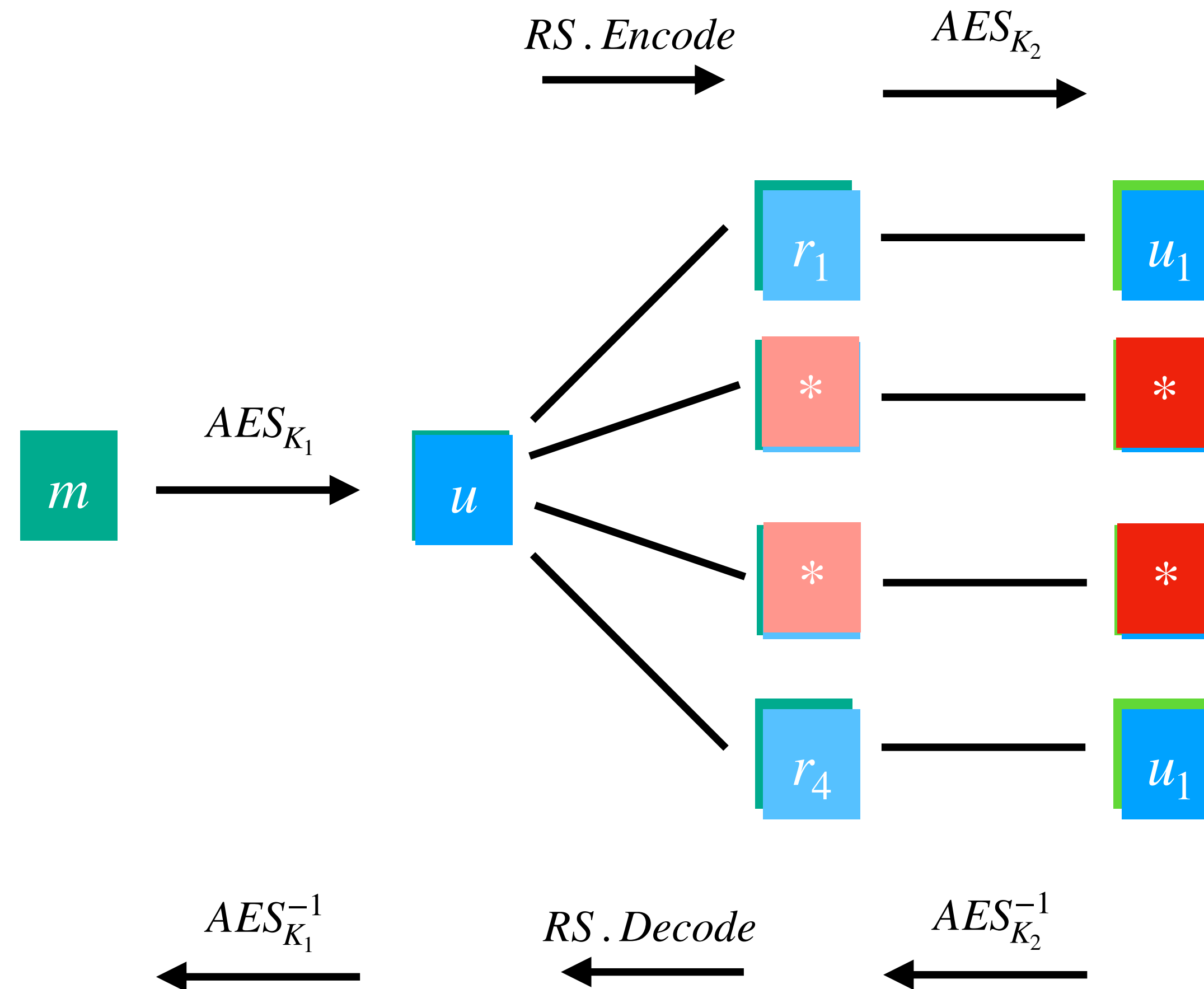
Pseudorandomness

$$m \leftarrow Dec(sk, cw + e)$$



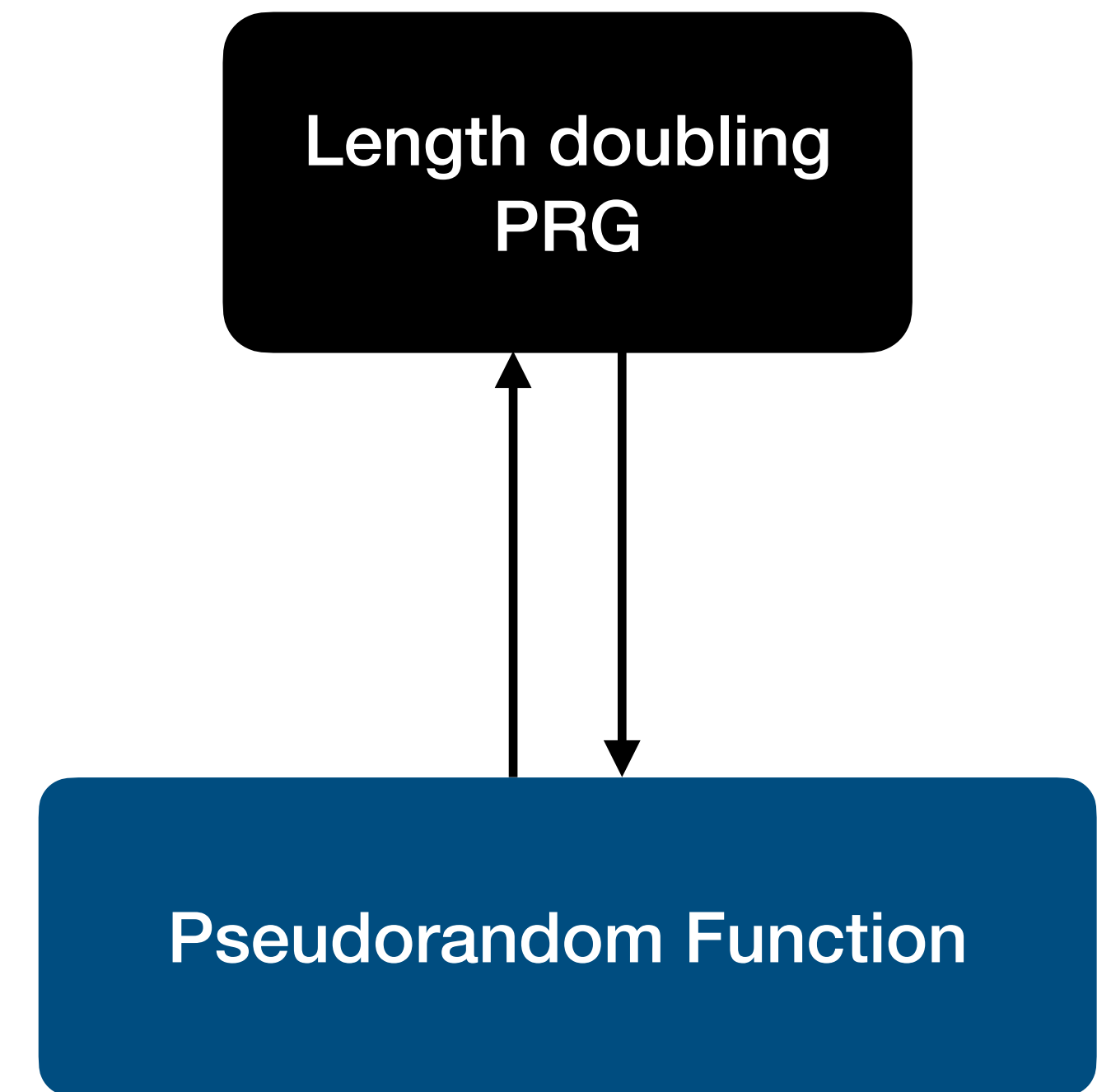
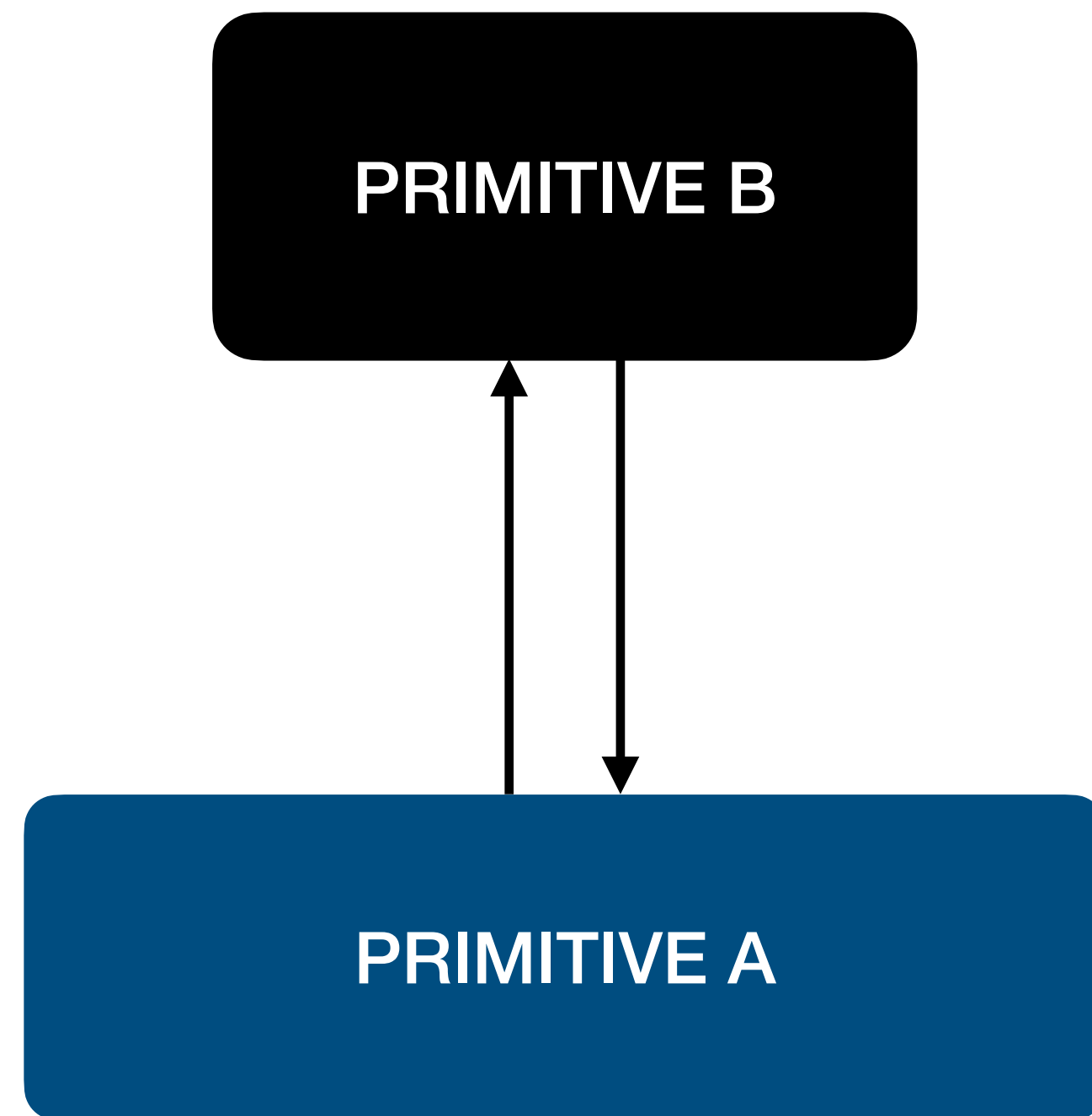
Robustness

Blackbox Construction for Large Alphabets



Impossibility Result

Blackbox Construction



Goldreich, Goldwasser, Micali'86

Our Result

*There is no blackbox construction of **pseudorandom codes** from*



One-Way Functions

One-Way Function

Any random function

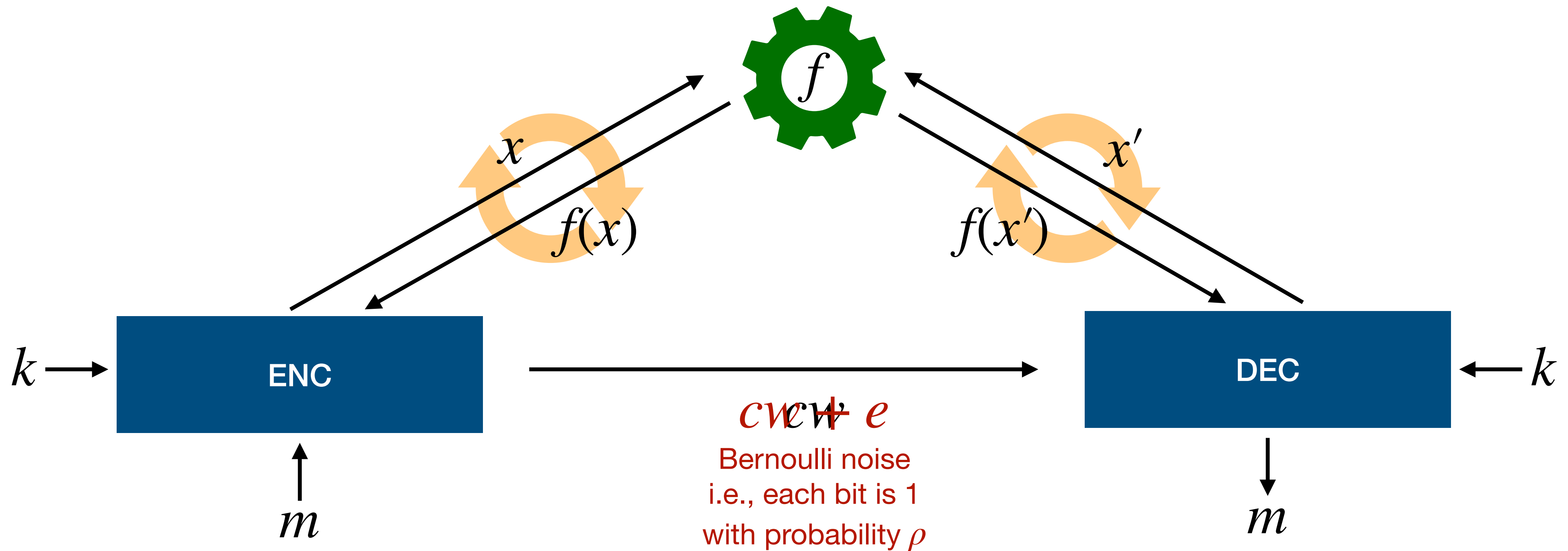
Multiply two large primes
Factoring $\approx \exp(O(\log(n)^{\frac{1}{3}} \log \log(n)^{\frac{2}{3}}))$

f

$x \xrightarrow{\text{Efficient}} f(x)$

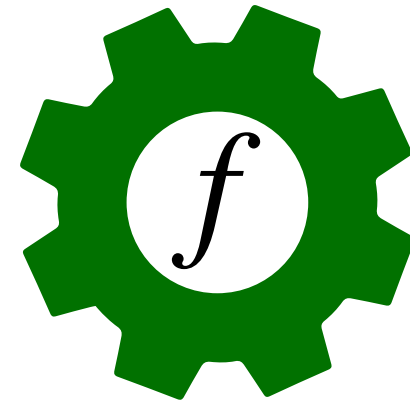
$r' \xleftarrow{\text{Hard to invert in polynomial time}} f(r)$
s.t. $f(r') = f(r)$ where r is random

Blackbox Construction of PRC from f



Security

But, only polynomially many queries allowed



Time and Space Unbounded

m

cw

I can only use f as a source of hardness

$b \leftarrow \{0,1\}$

$k \leftarrow \text{Setup}$

$\text{if}(b = 0) : cw \leftarrow \text{Enc}_k^f(m)$

$cw \leftarrow \text{random}$

Base Case



- $Dec_{(\cdot)}^f$ ^{makes} ~~does not make any~~ f queries that does not intersect with $Enc_{(\cdot)}^f$ queries !
- Sends random m_i to challenger and obtain corresponding cw_i .
- For each k , check whether $Dec_k(cw_i) = m_i$ for all i .
- If there exists such a k , output "PRC". Otherwise, "RANDOM".

Assumptions

$Dec_{(\cdot)}^f$ is deterministic.

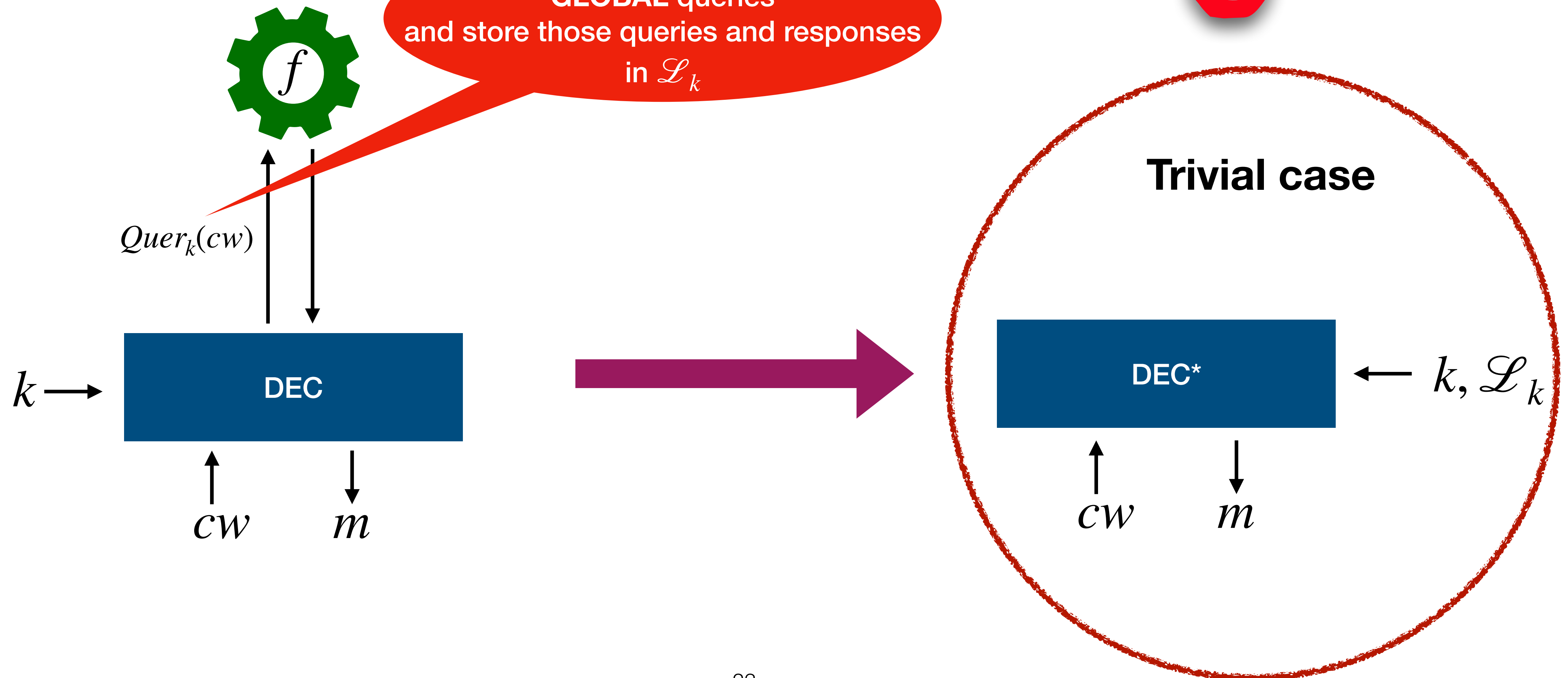
$Dec_{(\cdot)}^f$ makes a single f query.

High-Level Idea



I know DEC and
also $Quer_{(\cdot)}$

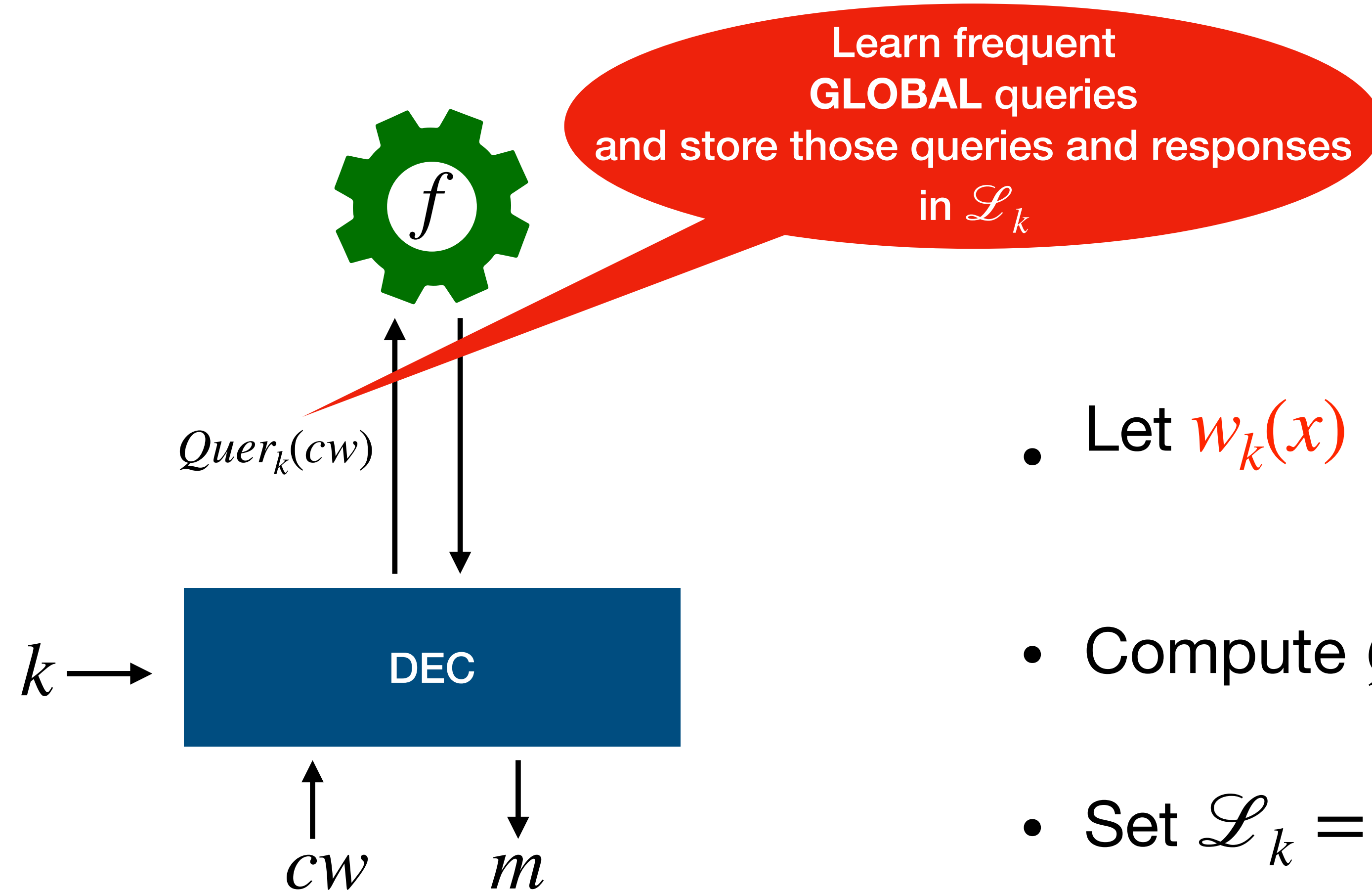
Learn frequent
GLOBAL queries
and store those queries and responses
in $\mathcal{L}_k$



Learning Global Queries

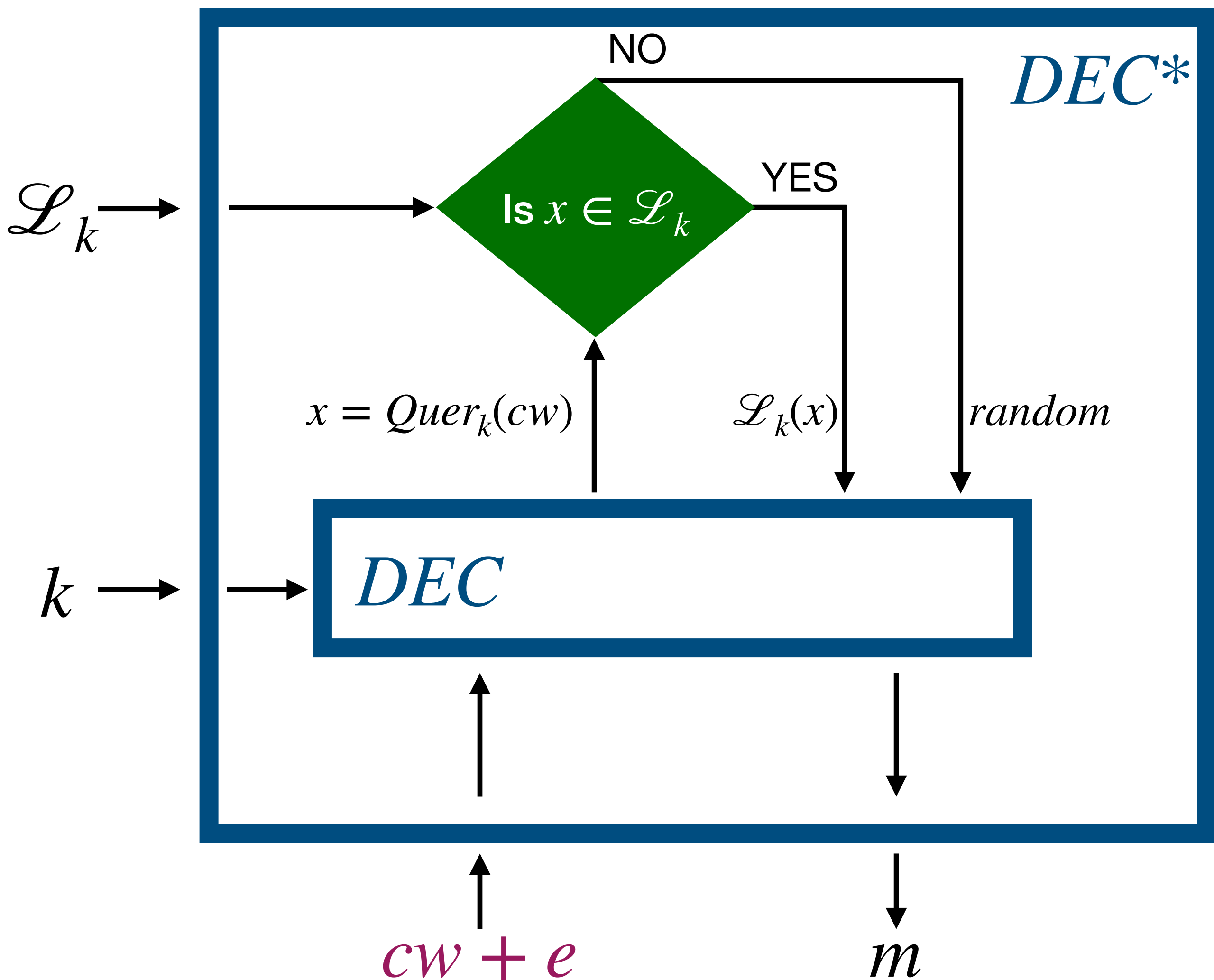


Time and Space
Unbounded



- Let $w_k(x) = \Pr_{u \leftarrow \{0,1\}^{|cw|}} [Quer_k(u) = x]$.
- Compute $\tilde{Q}_k = \{x \mid w_k(x) \geq 1/\text{poly}(\lambda)\}$.
- Set $\mathcal{L}_k = \{(x, f(x)) \mid \forall x \in \tilde{Q}\}$

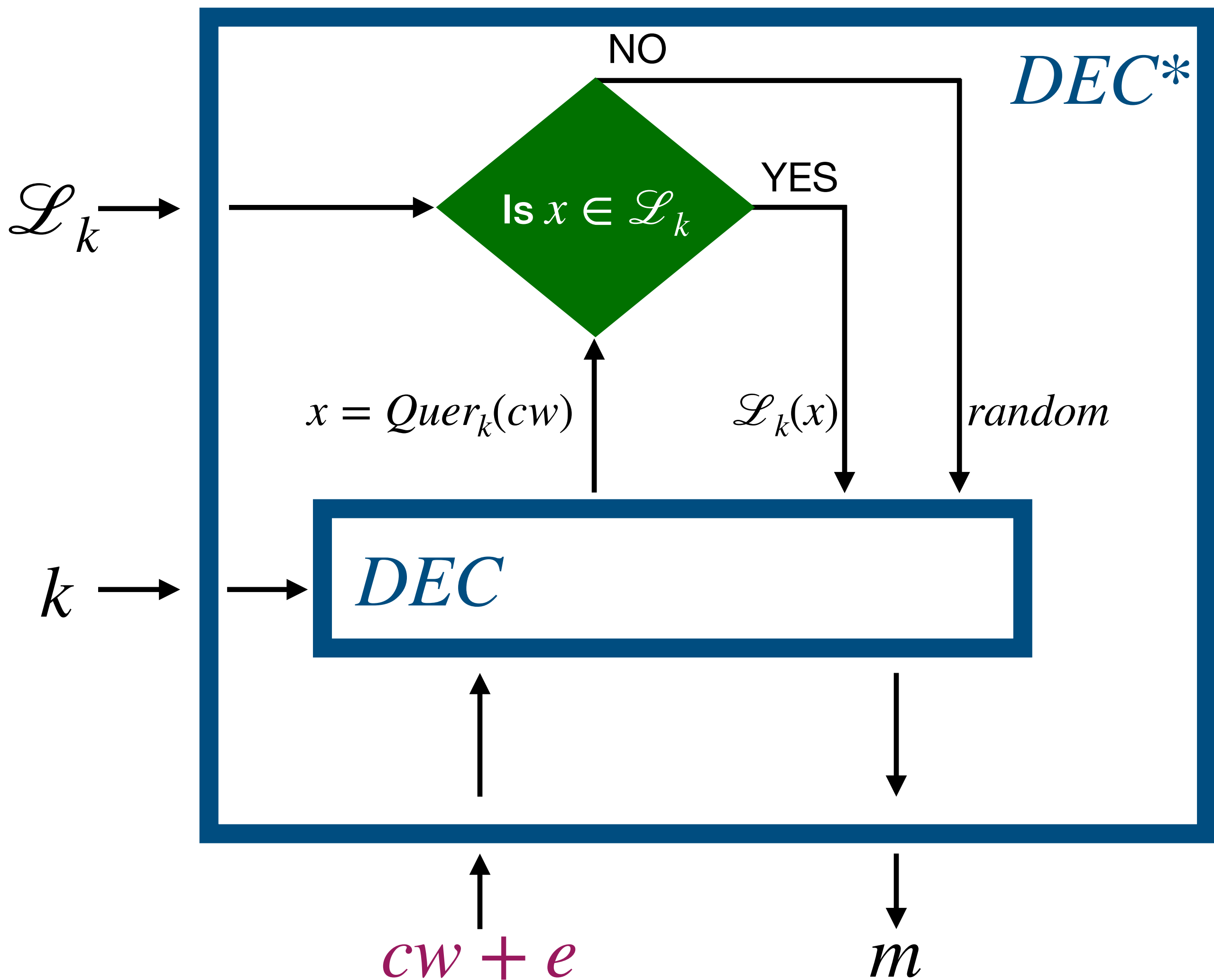
How does DEC^* work ?



- If “YES”, perfect simulation because $\mathcal{L}_k = \{(x, f(x))\}$.
- If “NO”, simulation might fail.

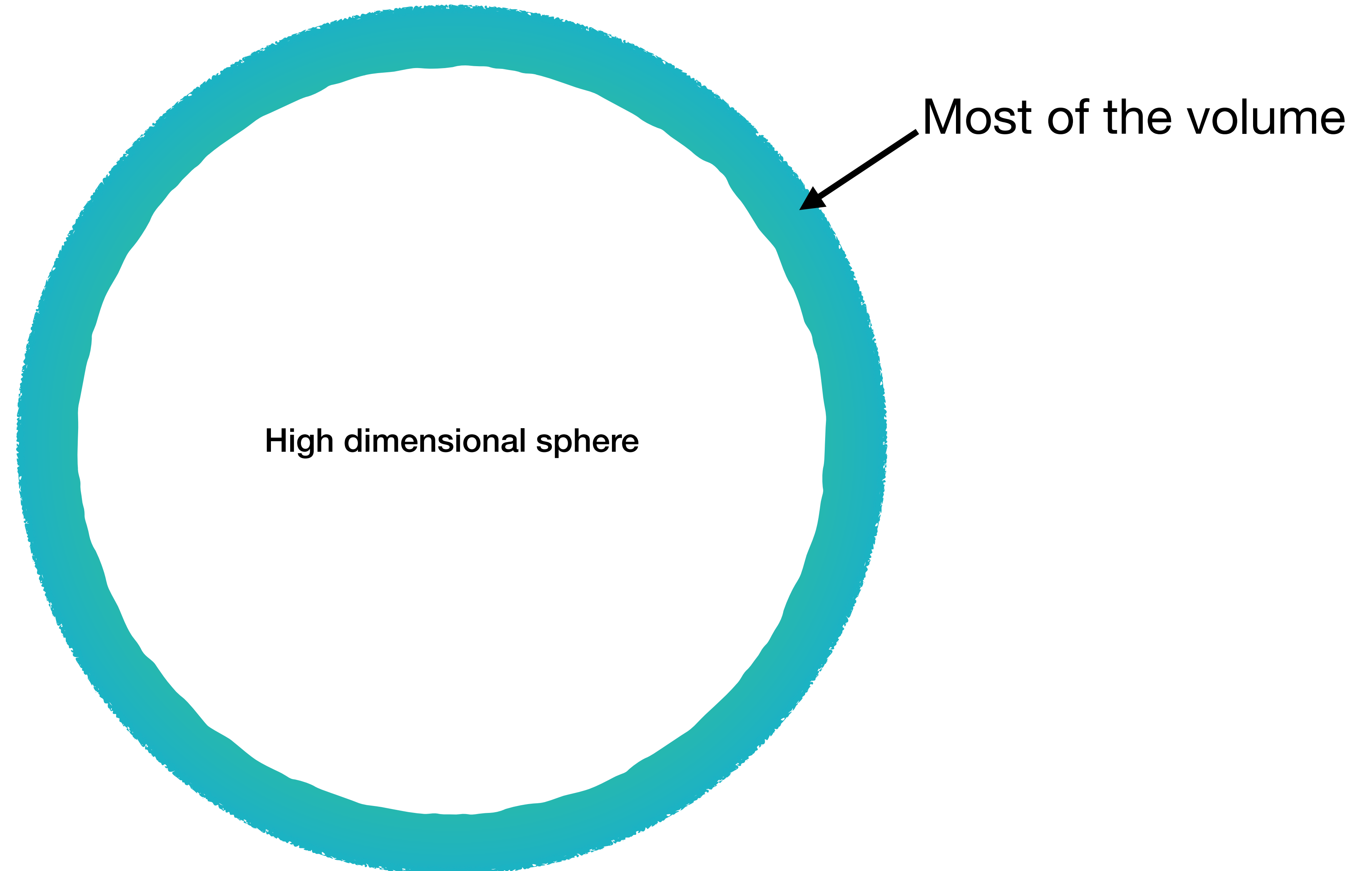
Are **NON-GLOBAL** but **LOCAL** queries frequent?

Worst-Case *Dec*

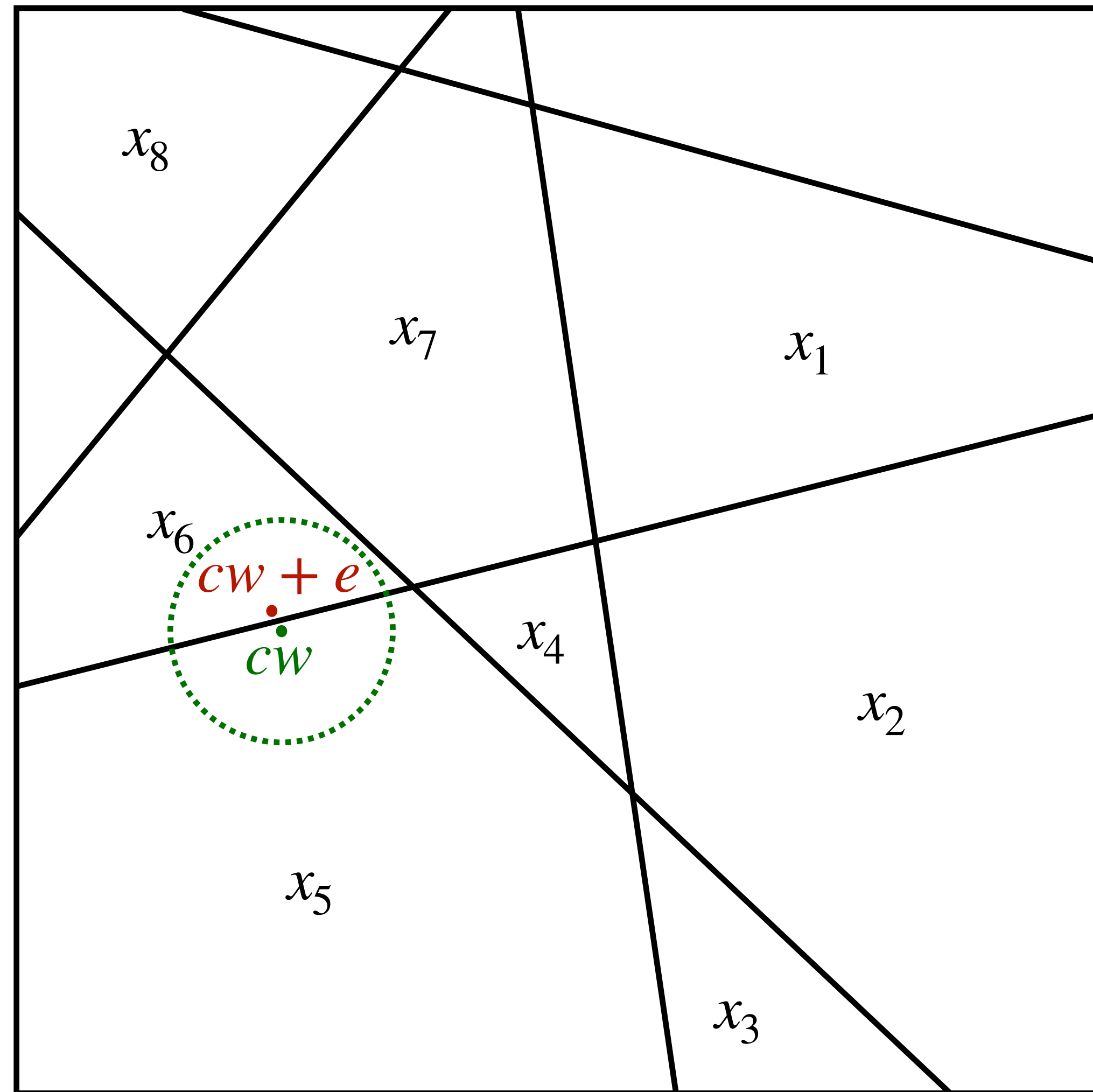


- For every codeword cw_i , $x_i = Quer_k(cw_i + e)$ is **unique**.
- Maybe possible that $w_k(x_i) = \text{negl}(\lambda)$
- In this case, “NO” is always executed!

Concentration of Measure



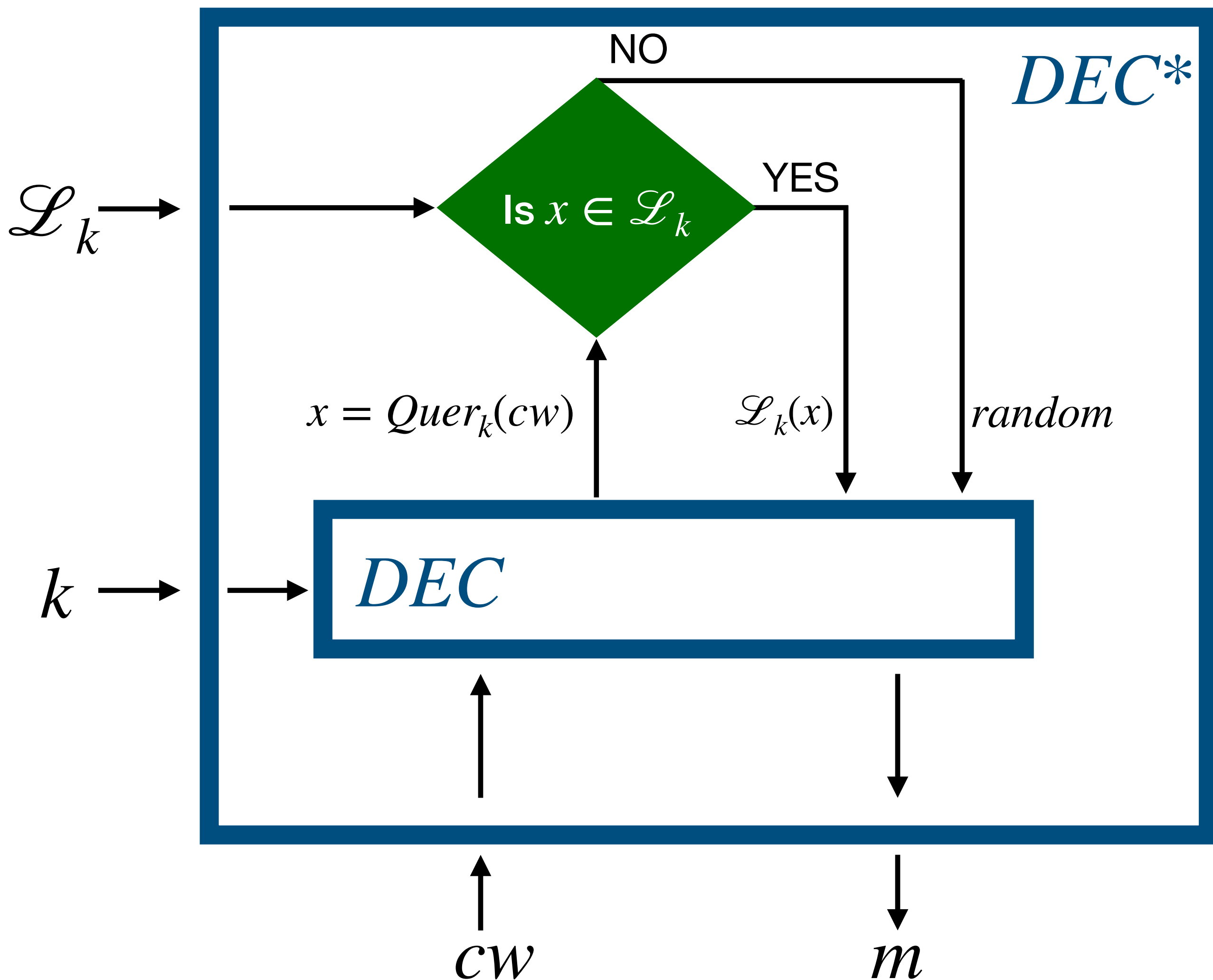
Hypercontractivity Theorem [Bon70][Bec75]



$$Quer_k : \{cw_i\} \rightarrow \{x_j\}$$

- Note that Dec_k must properly decode $cw + e$.
- If a Boolean function (including $Quer_k$) is not almost constant, then some input variable must have non-negligible influence on the output.
- **The theorem is for uniformly random strings.**
- Rely on pseudorandomness of the codewords.

How does DEC^* work ?



- If “YES”, perfect simulation because $\mathcal{L} = \{(x, f(x))\}$.
- If “NO”, simulation might fail.

NON-GLOBAL but LOCAL queries are not frequent !!!

Lifting to Multiple Queries

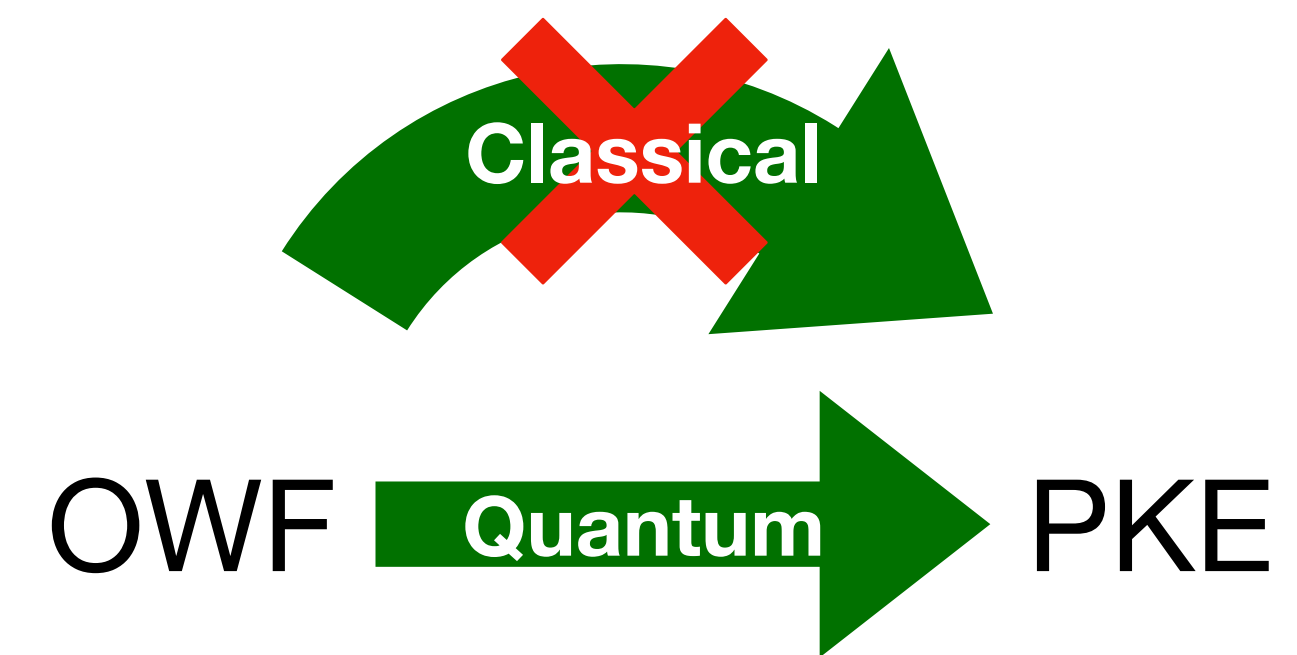
- Suppose $Dec_{(\cdot)}^f$ makes ℓ many queries.
- Perform the previous transformation with *Quer* being the **first query** made by $Dec_{(\cdot)}^f$.
- This will give $Dec_{(\cdot)}^{*f}$ that makes at most $\ell - 1$ many queries!

My Results

- Proved that PRCs **cannot be constructed** via black-box techniques from a broad class of standard cryptographic primitives — including **obfuscation**. [DMR-TCC'25, **Outstanding paper award**].
- Extended the **impossibility** result to the **quantum setting** also! [Under submission]
- Constructed PRCs secure against **stronger adversaries** (CCA security) using non black-box techniques [Accepted in CRYPTO 2026!].

Future Work on PRCs

- Develop **quantum** PRCs, where **codewords** are **quantum states**.
- Study decryption under different quantum **error models**
- Allow classical or **quantum** messages
- Identify minimal computational assumptions
- Explore more constructions/impossibilities for **plain** PRCs.



AI + Cryptography

Watermarking LLMs

Verifiable and private machine learning

Embedding backdoors in a model

Privacy: Model Stealing

And more ...



Thank You!

<https://mahe94.github.io>